H. Department of Defense Announced Possible Chemical Weapons at Khamisiyah: January - September 1996

Figure 39 shows the events from January to September leading to and immediately after the DoD's public announcement that the 37th Engineer Battalion probably destroyed bunkers containing chemical weapons.

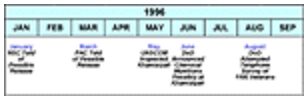

Figure 39. Investigation results through September 1996

CIA mentioned the possibility of agent release at Khamisiyah. In January, in a preliminary briefing to the National Security Council (NSC), the CIA mentioned the possibility of agent release at Khamisiyah.[181] The NSC directed the CIA and DoD to aggressively pursue this matter.

On March 5, the CIA informed a PAC staff member that US troops had been in the vicinity of a probable release of chemical warfare agent. On March 10, a CIA analyst heard a radio talk show on which a 37th Engineer Battalion veteran described demolition activities at a facility the analyst recognized as Khamisiyah. The CIA informed the PAC that week.

On May 1, at a PAC hearing in Washington, DC, the CIA and PGIT acknowledged that the 37th Engineer Battalion destroyed munitions at Khamisiyah and the agencies were working together to determine if chemical warfare agents were among the munitions destroyed.

In the spring of 1996, the PAC requested the CIA to examine the potential dispersion of nerve agent from the March 1991 demolition of Bunker 73 at Khamisiyah and at two other sites in Iraq (Al Muthanna and Muhammadiyat).[182,183] Among other models, the CIA used the US Army's Chemical and Biological Defense Command's Non-uniform Simple Surface Evaporations 4 (NUSSE4) transport and diffusion model.[184]

On May 14, UNSCOM inspectors documented the presence of burster tubes, fill plugs, plastic inserts, and other items characteristic of chemical weapons in Bunker 73 at Khamisiyah. Iraq reiterated its declaration that US forces had destroyed chemical munitions in Bunker 73, and for the first time claimed that US forces destroyed chemical munitions stored in the Pit.[185]

DoD publicly announced US forces had probably destroyed bunkers containing chemical weapons

> UNSCOM has informed us that, as part of its ongoing effort to verify Iraqi declarations, it inspected the Khamisiyah ammunition storage area last month [May 1996]. During that inspection, UNSCOM concluded that one bunker had contained rockets with chemical agents. US soldiers from the 37th Engineer Battalion destroyed ammunition bunkers at this site in early March 1991, shortly after the war ended. Based on a new review of the available information, it now appears that one of these destroyed bunkers contained chemical weapons.[186]

After the June 21, 1996, announcement, the investigation's focus shifted to better understand two questions:

- First, what was the potential for exposure to chemical agents from the demolition operations at Khamisiyah?
- Second, who might have been exposed?

PGIT began interviewing US soldiers directly involved in the demolition of Bunker 73 to reconstruct such information as the exact dates of the demolition, amount and type of munitions destroyed, and weather and wind

direction on the dates of demolition. This information was provided to the CIA to assist in their dispersion modeling of the Bunker 73 demolition.

In July, the CIA briefed the results of its Bunker 73 modeling effort to the PAC and, on August 2, 1996, published their report, "Report on Intelligence Related to Gulf War Illnesses," which identified their modeling assumptions and concluded that any hazard area resulting from the demolition of Bunker 73 moved east and northeast. Among the CIA's more significant assumptions in modeling the demolition of Bunker 73 were:

- Bunker 73 contained approximately 1,060 nerve agent-filled rockets;
- Each rocket was filled with 8 kilograms of a 2:1 ratio of sarin to cyclosarin nerve agent;
- The demolition ejected 10 percent of the rockets from the bunker;
- Based on US tests, heat from the explosion and burning motors and crates degraded 2.5 percent of the agent in the bunker;
- Winds were light to the northeast to east; and
- The modeling did not include the effect of thermal energy released by the simultaneous burning and detonation of the other 32 to 37 bunkers.[187]

In August and September DoD attempted a telephone survey of Khamisiyah participants. Because of the uncertainty whether US forces had been exposed to chemical warfare agents, the Assistant Secretary of Defense for Health Affairs designed and conducted a telephone outreach program to contact veterans who may have participated in the Khamisiyah operation. The Defense Manpower Data Center (DMDC) attempted to contact more than 1,100 veterans of units PGIT had determined were in the Khamisiyah area during early March 1991. The DMDC was able to contact 575 of them and asked them to call the DoD hotline to report any medical problems they were experiencing and provide any information they believed pertinent to the Khamisiyah incident. Deputy Secretary of Defense White wrote the approximately 525 veterans not contacted by telephone to urge them to call the Persian Gulf Incident Hotline. This particular telephone outreach effort concluded in October 1996.

The Secretary of Defense widened DoD's investigation. In September 1996, DoD dramatically increased its investigative efforts. As announced in a news release, Secretary of Defense White:

- Ordered the establishment of a DoD Persian Gulf Illness Action Team "to look across all the activities ongoing in the Department related to Persian Gulf veterans' illnesses, reassess whether we're in the right direction with the right amount of resources and level of effort, and asking the right questions."
- Directed the Army to conduct an Inspector General inquiry into events surrounding Khamisiyah and supplement the efforts of the DoD investigation where possible.
- Directed the Assistant to the Secretary of Defense for Intelligence Oversight to investigate US intelligence on Khamisiyah, including how the information was handled.[188]

On October 2, 1996, the Secretary of Defense named Dr. Bernard D. Rostker, then Assistant Secretary of the Navy for Manpower and Reserve Affairs, to head an action team to evaluate the avtivities of the DoD related to undiagnosed illnesses of Gulf War veterans. He was to report his recommendations to the Deputy Secretary of Defense."[189] The Office of the Special Assistant assumed the team functions one month later.

I. Determining Possible Troop Exposure: October 1996 - December 1997

Efforts continued from fall 1996 to late summer 1997 to resolve uncertainties and complete the initial modeling effort for the demolition at the Pit (Figure 40). Investigators had determined even less about the March 10, 1991, demolition in the Pit than they had about the March 4, 1991, destruction of Bunker 73. DoD and the CIA jointly continued to investigate the activities that occurred in the Pit. Critical uncertainties persisted about such facts as weather, amount and placement of charges, number of participants, and number of events. Accurately modeling the effects of the Pit demolition would be impossible without resolving these uncertainties. Also, since the two agencies knew little about how such an explosion would react, they decided they needed to conduct field tests to resolve some of these questions. Thus, DoD and the CIA delayed modeling the effects of the Pit demolition until

the investigators could develop more information from ongoing veterans' interviews, document research, and field tests.

Figure 40. Actions taken to resolve uncertainties and conlcude initial modeling effort

Fall 1996

DoD announced its intent to survey US troop units within 50 kilometers of the Khamisiyah ASP. After reviewing the CIA's preliminary work to model the Pit demolition, DoD had considerable uncertainty concerning the fallout from the March 10, 1991, demolition. Therefore, in October 1996, DoD announced it would survey the estimated 20,000 veterans who had ben in units within 50 kilometers of the Khamisiyah ASP during the period March 1 - 15, 1991, according to a unit locator database maintained by ESG, now the US Armed Services Center for Unit Records Research (USASCURR).[190] The survey would be an attachment included in Deputy Secretary of Defense White's letter to the veterans that would indicate that chemical weapons had been present at Khamisiyah when the demolitions occurred. The letter urged veterans to call the Persian Gulf Incident Hotline with any additional information about the Khamisiyah incident or to report illnesses they attributed to their service in the Gulf War. The Deputy Secretary of Defense did not send the letter to the 1,100 veterans previously identified for the telephone survey. In preparing the survey distribution, DoD carefully selected the dates and distances to be sure to identify and notify units moving through the area between March 1 and 15. The potential exposure assumptions translated into three unit location zones for the survey participants:

- A 5-kilometer zone enclosing the area where chemical warfare agent exposure would have been lethal to any inhabitant;
- A 25-kilometer zone encompassing the area where chemical warfare agent exposure would have produced first noticeable effects, such as runny nose and blurry eyes; and
- A 50-kilometer zone containing an area that doubled the 25-kilometer zone as an added safety measure to ensure including in the survey all US forces in transit through the Khamisiyah area.

DoD also selected the wider range of transit dates for the veterans' locations because conflicting information existed about the number and dates of the demolitions. An EOD log, dated March 12, 1991, had an entry of a possible demolition similar to the confirmed demolition of March 10, 1991. Since analysts were investigating the conflicting dates, DoD added three days following March 12 as a precaution, making the survey period March 1 - 15.[191]

The Institute for Defense Analyses recommended numerous changes in modeling the Pit. At DoD's and CIA's request, the Institute for Defense Analyses (IDA) convened an independent panel of experts in meteorology, physics, chemistry, and related disciplines. The panel reviewed CIA's modeling methodology and analysis, which used the analytical linkage between the Defense Threat Reduction Agency's Operational Multi-scale Environmental Model with Grid Adaptivity (OMEGA) weather model and the Naval Surface Warfare Center's Vapor, Liquid, and Solid Tracking (VLSTRACK) dispersion model to drive the Army Chemical and Biological Defense Command's NUSSE4 transport and diffusion model.[192] The IDA recommended using additional atmospheric models and data sources for modeling the demolition in the Pit.[193]

DoD established the Office of the Special Assistant for Gulf War Illnesses. On November 12, 1996, at the recommendation of the Persian Gulf illnesses action team, the Deputy Secretary of Defense created this office and named Dr. Bernard D. Rostker as the Special Assistant. He assumed responsibility for the PGIT, which was incorporated as the office's Investigation and Analysis Directorate.[194]

1997

The Special Assistant for Gulf War Illnesses sent the Khamisiyah survey. The Special Assistant for Gulf War Illnesses sent the survey, dated January 1997, accompanied by a letter of explanation, to veterans whose units we determined from the then ESG unit locator database were to have been within the 50-kilometer circle around the Khamisiyah ASP between March 1 and March 15, 1991.[195]

President extended the PAC's tenure. On January 30, 1997, the President extended the PAC's tenure until October 31, 1997, with tasking to provide a Supplemental Letter Report and Supplemental Final Report, the PAC's "Special Report."

The Presidential Advisory Committee on Gulf War Veterans' Illnesses provided supplemental letter report. On April 30, 1997, the Presidential Advisory Committee on Gulf War Veterans' Illnesses provided details of its work in two areas: continuing oversight of the government's investigation into Gulf War chemical and biological warfare incidents and implementing recommendations of the PAC's Final Report. Reiterating its January 7, 1997, "Final Report," the PAC stated, "In the face of credible evidence of the presence or release of chemical warfare agents, low-level exposure must be presumed while efforts to develop more precise measures of exposure continue….Troops within the presumptive exposure area should be notified and encouraged to enroll in the CCEP or [VA] Registry." Hence the Committee noted:

> DOD should move as quickly as possible toward conclusions about the incidents under investigation and, when in doubt, err in favor of targeted notification of troops about possible health risks and the availability of free diagnosis and treatment programs established by the government.[196]

Office of the Special Assistant for Gulf War Illnesses and the CIA continued cooperative investigations to reduce the uncertainties concerning demolition activities in the Pit. In his April 24, 1997, opening statement to the House Committee on Government Reform and Oversight Subcommittee on Human Resources and Intergovernmental Relations, Robert D Walpole, Special Assistant to the Acting Director, CIA, stated:

> However, when we turned to modeling demolitions at the Pit, we quickly realized we had significant uncertainties regarding how rockets with chemical warheads would have been affected by open-pit demolitions. We were also uncertain about the number of demolition events and the weather conditions at the time of the demolitions... CIA and DoD have devised a joint plan which will reduce some of these uncertainties in order to more accurately identify the extent of the release. [197]

Subsequently, DoD and CIA worked aggressively to eliminate as many as possible of these uncertainties surrounding the events in the Pit:

- Very limited, often contradictory information from several soldiers;
- Questions on the demolition date(s);
- Uncertainties on the number of rockets, agent purity, and amount of agent aerosolized;
- Uncertainties on agent release and release mechanisms in an open-pit demolition;
- Limited weather data; and
- No single model that simultaneously ran weather and chemical warfare agent data.[198]

Interviews with soldiers reduced uncertainties. We located and, with the CIA, interviewed five soldiers who had key roles in the demolition activities in the Pit. These individuals included the 37th Engineer Battalion operations officer, who initially discovered the Pit and led engineers and EOD personnel to it; two 37th Engineer Battalion soldiers who placed demolition explosives on the stacks of rockets; the 60th EOD Detachment soldier

who supervised the engineers;[199,200] and the 60th EOD Detachment executive officer who inspected the stacks before detonation.[201] They confirmed the number of stacks of rockets wired for demolition and other information, such as the amount and placement of explosives on the rockets, that proved invaluable to the DoD/CIA team's field tests discussed in later paragraphs.

Soldier interviews clearly established two large-scale demolitions. An entry in a 60th EOD detachment field log had indicated a third demolition on March 12. Interviews of 60th EOD Detachment soldiers revealed the date was incorrect.[202] This clarification allowed the DoD/CIA team that would model the event to concentrate on one major explosion in the Pit and not two separate events.

The DoD/CIA team estimated the number of 122 millimeter rockets in the Pit to be 1,250. In May 1996, Iraq had declared to UNSCOM that 1,100 rockets were in the Pit at the demolition. The DoD/CIA team used personal interviews, estimates of the heights of the stacks of 122mm rockets, and known data, such as the size of the rocket crates, to develop the estimate of 1,250 122mm rockets. About six months after the demolition, UNSCOM found about 750 of the rockets still contained chemical warfare agent. UNSCOM assisted Iraq in disposing of these rockets. Therefore, 1997 assessments assumed that the demolition destroyed approximately 500 rockets on March 10, 1991.[203]

Test team determined the rockets' nerve agent capacity. In preparing for the May field demolition testing on the rockets, the test team determined that a single rocket would hold only 6.3 kilograms of agent instead of the 8 kilograms used in the modeling of Bunker 73.[204] The previous estimate failed to account for the payload volume reduction caused by the presence of two plastic canister inserts whose total weight was 1.7 kilograms.

The DoD/CIA team estimated chemical agent ratio and purity. The DoD/CIA team estimated chemical agent purity at 50 percent and the ratio of sarin/cyclosarin to be 3:1, based on a combination of estimates taken from UNSCOM samples, Iraq's chemical production records, and Iraq's declarations. In their "Modeling the Chemical Warfare Agent Release at the Khamisiyah Pit" document, DoD/CIA stated:

> Our best estimate of the agent purity at the time of the demolition is slightly less than 50 percent. Iraqi production records obtained by UNSCOM indicated that the sarin/cyclosarin (GB/GF) nerve agent produced and transported to Khamisiyah in early January 1991 was about 55 percent pure. The agent, subsequently, degraded to 10 percent purity by the time laboratory analysis had been completed on samples taken by UNSCOM from one of the rockets in October. On the basis of the sample purity and indications that the degradation rate for sarin and cyclosarin are similar, we assess that the ratio when the munitions were blown up in March 1991, was the same as that sampled in October 1991 - 3:1. Assuming a conservative, exponential degradation of the sarin/cyclosarin, the purity on the date of demolition two months after production can be calculated to be about 50 percent.[205]

The DoD/CIA team confirmed initial wind direction from existing records and photography. The DoD/CIA team used existing data and photography to determine the initial wind direction on March 10, 1991. Very little weather data was readily available for March 10th, so the team determined the weather by combining exact location coordinates of Khamisiyah, general weather conditions during March 1991, imagery from March 10 - 11, photography of soot patterns created by the bunker demolition on March 10th, and regional scale imagery showing the Kuwait oil field fire plumes for the days immediately following the demolitions. Using these sources, the DoD/CIA team determined the initial wind direction at Khamisiyah to be from the north-northwest, blowing any chemical release to the south-southeast.[206]

The DoD/CIA team conducted field demolition tests. Since the CIA used test information that did not apply to open-air demolitions to model Bunker 73, the test information could not be used to determine if agent would

release quickly or would take a period of days. By April, the DoD/CIA team agreed that field demolition tests were necessary[207] to determine how chemical agents would behave in an open area similar to the Pit. The team arranged to conduct a series of tests at Dugway Proving Ground, Utah from May 15 to 31, 1997. The DoD/CIA design team developed the Dugway tests (Figure 41) to extrapolate the interaction between the explosives and rockets in an open environment similar to the Pit demolition. The tests determined:

- How the rockets reacted to explosives placed in various locations on them;
- How the nerve agent was released (e.g., aerosolized amount and evaporation rates of spilled agent on soil and wood); and
- How much nerve agent was released.[208]

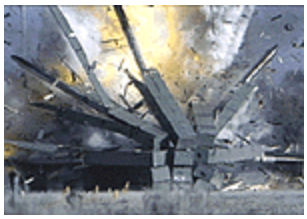

Figure 41. Demolition test; picture courtesy of Dugway Proving Ground

For a thorough evaluation of the results of those tests, see "Modeling the Chemical Warfare Agent Release at the Khamisiyah Pit," September 4, 1997.[209]

The Office of the Special Assistant continued efforts to identify troop unit locations. The Special Assistant sent the Khamisiyah survey to veterans identified through an existing database containing unit locations called the Persian Gulf Registry. USASCURR, formerly the Environmental Support Group, derived Gulf War unit locations by reviewing a large number of Gulf War unit records, including unit history data archives, operational logs, situation reports, and after action and historical reports. USASCURR began this review in mid-1994 and has repeatedly updated the database as it obtained additional unit locations.

USASCURR continued recording unit location data in conjunction with the Gulf War declassification initiative the Deputy Secretary of Defense established in March 1995. The declassification initiative decreed a DoD-wide effort to review Gulf War operational records, declassify them, and make them available to USASCURR and the Office of the Special Assistant.

To identify veterans who may have experienced low-level chemical exposure, we determined their units' locations from the USASCURR unit location database during March 10 - 13, 1991. Although veterans were either assigned or attached to specific units during the Gulf War, a unit's location on a specific day did not necessarily pinpoint where an individual soldier was on that day. For example, a precise location recording a soldier on patrol or in transit to another location does not exist.

In April 1997, the Office of the Special Assistant and Department of the Army began a coordinated effort to assemble former Gulf War brigade, divisional, and non-divisional units operations officers (G3s and S3s) to verify existing locations in the USASCURR unit location database and assist in identifying additional unit locations.

The DoD and CIA linked models to determine the downwind hazard area of the March 10, 1991, Pit demolition. To meet the IDA panel's recommendations, the DoD/CIA asked other agencies with long-time modeling experience to participate in the modeling effort. The modeling team consisted of scientists from the Defense Special Warfare Agency (DSWA) (now the Defense Threat Reduction Agency (DTRA)); the Naval Research Laboratory (NRL); Naval Surface Warfare Center (NSWC); and National Center for Atmospheric Research (NCAR); and Science Applications International Corporation (SAIC) supporting the CIA and DSWA. These agencies used existing sophisticated models to develop Khamisiyah-specific potential exposure areas. As IDA recommended, the team used different combinations of models to reduce model bias. Presenting a

composite (union) of the different modeling simulations representing the overlay of the outermost perimeter of all models determines the hazard area.[210] A general discussion of the modeling process is at Tab E.

The DoD/CIA team announced no US troops were in the area predicted to have noticeable health effects during the time of the event. On July 24, 1997, DoD/CIA team announced the results of the potential hazard area modeling effort.[211] The results confirmed no US units were within the hazard area predicted to cause noticeable short-term health effects from demolition activity in the Pit. However, the modeling results did indicate that troops in Iraq and Saudi Arabia possibly were exposed to low levels of nerve agent over a four-day period from March 10 - 13, 1991. Using data on then-available unit locations, DoD identified 98,910 soldiers within the potential hazard area predicted by the models.[212] From late July through September DoD sent written notices to two categories of veterans: those in the potential chemical agent hazard area (approximately 99,000) and those who had received the Deputy Secretary of Defense's letter and survey but were not in the potential chemical agent hazard area (approximately 10,000). On September 4, 1997, the DoD/CIA team published the details of this modeling effort in the document: "Modeling the Chemical Warfare Agent Release at the Khamisiyah Pit."

In response to the announcement of the modeling results in July 1997, various scientific groups recommended refinements to the models and modeling process. After addressing these groups' refinements, the team decided to generate a new set of Khamisiyah simulations. We have completed these simulations and are publishing the results in this paper (see Remodeling Pit Demolitions section) and in more detail later in a technical report.

The PAC issued Its Special Report. The PAC issued a Special Report on October 31, 1997, to the Secretary of Defense, and its service ended in November 1997.

J. Remodeling Effort: January 1998 - March 2000

Figure 42. Activities surrounding the remodeling of the Khamisiyah Pit demolition

1998 - 1999

CIA released Inspector General report. On February 5, 1998, the CIA Inspector General released its assessment of the CIA's analysis and reporting of Khamisiyah information from the mid-80s, when Khamisiyah was first identified as a possible chemical weapons storage site, to June 1996, when DoD announced US forces had destroyed bunkers possibly containing chemical weapons. The Inspector General came to two conclusions:

- Information obtained before, during, and after the Gulf War, was not compiled and comprehensively analyzed until the US government focused on Gulf War illnesses in 1995.
- A more vigorous follow-up when questions arose about Khamisiyah might have brought about an earlier realization that US troops may have been exposed to chemical agents.[213]

Senator Rudman appointed as Chair of Presidential Special Oversight Board. On February 24, 1998, President Clinton named Senator Warren B. Rudman as Chair of the Special Oversight Board for Department of Defense Investigations of Gulf War Chemical and Biological Incidents (PSOB). President Clinton established the PSOB by Executive Order to provide recommendations based on its review of DoD's investigation into possible Gulf War chemical and biological incidents that may have contributed to Gulf War veterans' illnesses. The board reports to the President through the Secretary of Defense.[214] Following appointment, the full board met for the first time in July 1998 and received its first briefing from DoD on the history and background of Gulf War issues. The PSOB continues to review and approve the overall direction of the Special Assistant's office.[215]

G3/S3 conferences concluded. From April 1997 through June 1998, the Office of the Special Assistant and the Department of the Army brought 163 former Gulf War corps, division, and brigade operations officers (G3/S3) to USASCURR to identify unit locations and verify their respective Gulf War units existing locations. Successively, Army operations officers from XVIII Airborne Corps, VII Corps, and Echelons-Above-Corps units returned to assist in this effort, which significantly enhanced KTO unit location information. Army operations officers added approximately 390,000 locations to the USASCURR database, more than doubling the Army locations known before the G3/S3 conferences. Currently the number of locations in the database for all services during the Gulf War period is more than 855,000. Tab F contains details of the development and results of the G3/S3 conferences.

Senate Veterans' Affairs Committee report released. The Senate Veterans Affairs Committee released its Special Investigation Unit's report on Gulf War illnesses in August 1998. While the report complimented the Office of the Special Assistant for its direct approach in dealing with veterans, it also pointed out areas requiring improvement in both the DoD and VA. In addition, the Special Investigative Unit acknowledged the value of the Office:

> Establishment of the Office of the Special Assistant for Gulf War Illnesses in 1996 has increased the flow of information to veterans and the public about various events during the Gulf War that may have affected the health of veterans who served there. OSAGWI has also made efforts to solicit from Gulf War veterans their concerns about their health and possible exposures, and should continue these efforts.[216]

Panel recommended Remodeling Pit demolition. Following the 1997 Khamisiyah pit demolition modeling, a panel of modeling experts evaluated the modeling results. This panel approved the DoD/CIA modeling methodology but recommended a number of improvements including revisions to the computer models used. The Office of the Special Assistant initiated improvements to the 1997 model process to obtain the highest quality of hazard area definition possible. Modeling improvements continued throughout 1998 and 1999 and culminated in redefined potential hazard areas in January 2000.

2000

Independent peer review panel responded. The Office of the Special Assistant's February 23, 2000, report, "Methodology of Refined Modeling of the Khamisiyah Pit Demolition," to an independent peer review panel described the revised modeling methodology and improvements. The panel found the revised modeling procedures satisfactory but stated modeling results would "still be on the conservative side; i.e., they are very likely overestimates of the dosages actually received by personnel."[217] As we did in 1997, our 2000 modeling approach overlaid the outer boundaries of the various modeling results to develop the composite potential hazard area. Our fundamental modeling approach has not changed since 1997:

> Remember that this plume [potential hazard area] is the composite of five models; the plumes from each individual model predicted smaller exposure areas. We used the composite approach to increase our confidence that the resulting plume would be our best estimate of the potential area covered, taking into account individual model biases. This approach was critical for notifications and for future epidemiological studies. However, we do not expect that everyone under the composite plume was exposed.[218]

We recognize that our composite approach methodology overstates veterans' risk of exposure. Each of the four Khamisiyah model combinations yielded a distinct potential hazard area. However, we cannot say with any degree of certainty which particular model's potential hazard area or which combination of potential hazard areas correctly identifies military units that may have been exposed. Hence, we used a composite of all four model combinations to reduce the possibility of missing potentially exposed military units. With this in mind, our goal remains the same as it was in 1997: if we err, we do so on the side of veteran notification. This is

particularly important for those veterans who will receive notice of possible exposure to low levels of nerve agent. (See Tabs E and F.)

Modeling improvements. The 2000 modeling improvements include the following:

- **Revised meterological models.** Meterological model changes were made in: (1) modeling options and input data options for two models: Coupled Ocean-Atmospheric Mesoscale and Prediction System (COAMPS) and Mesoscale Model, Version 5 (MM5), and (2) modeling codes for Operational Multiscale Environmental Model with Grid Adaptivity (OMEGA).
- **Reduced amount of nerve agent released.** On October 14, 1999, the Special Assistant to the Director of Central Intelligence for Persian Gulf War Illnesses Issues provided revised nerve agent release estimates for several Gulf War incidents to the Special Assistant. All-source intelligence information supported by UNSCOM data generated the revised nerve agent release estimates for number of rockets, quantity of agent, agent purity, and amount of agent aerosolized and evaporated for use in any future ensemble modeling. The CIA revised the Khamisiyah Pit source term data from the estimate used in their 1997 modeling. The CIA also revised source term data for the Bunker 73 although we did not remodel Bunker 73 in the 2000 effort. The revised source term estimates, quoted from the CIA letter, are:

  > Khamisiyah Pit - Worst case estimate of agent release: 321 kg GB/GF [sarin /cyclosarin] with the added comment: "New value is 45 percent of the previous assessment based on re-evaluation of number of rockets stemming from UNSCOM excavation in 1998. *The new lower release amount and inclusion of environmental degradation* [which this document calls "chemical warfare agent decay"] *will change exposure limits.*"
  >
  > Khamisiyah Bunker "73" - Worst case estimate of agent release: 51 kg GB/GF [sarin/cyclosarin] with the comment: "New value is lower based on reassessments of rockets and agent fill stemming from UNSCOM excavation in 1998. *Winds blow away from troops and release is five percent of release modeled in 1996 that did not expose troops.*"[219]

  The revised 2000 modeling indicated a release of 321 kilograms of nerve agent from the Pit instead of the 1997 release of 715 kilograms (a 55 percent reduction.) Likewise, the 2000 modeling revised the total as 51 kilograms instead of the 1997 figure of 1,040 kilograms of nerve agent released from Bunker 73.

  According to the October 14, 1999, CIA estimate, the number of 122mm rockets damaged during the Khamisiyah Pit demolition was 225, instead of the previously estimated 500. Thus, the 2000 source term was 45 percent of the 1997 source term.

- **Combined toxicity of sarin and cyclosarin included.** The 1997 model treated the source term as totally sarin (GB) because toxicity data on cyclosarin (GF) was not available then. In the 2000 model, we modeled the toxicity of both nerve agents using newly determined cyclosarin toxicity data. Although the amount of sarin was three times that of cyclosarin, the toxicity of cyclosarin is nearly three times that of sarin. Equally important, cyclosarin is less volatile than sarin (i.e., it evaporates more slowly). Modeling the toxicity of both nerve agents produced more accurate hazard areas. As mentioned, the 2000 estimate of agent released was less than half the 1997 estimate. However, the 2000 hazard area did not correspondingly decrease to half the size because of the relative increased toxicity of GF.[220]
- **Atmospheric Removal Mechanisms - deposition and degradation included.** We included deposition, i.e., settling to the surface, and degradation, often used synonymously with "decay," in the 2000 model. Both agent removal mechanisms more accurately depict the potential hazard area. Tab E explains in more detail the effects of removal mechanisms on the modeling process.
- **Updated unit location and personnel data used.** The USASCURR, with more than 855,000 unit locations in its database, maintains two data sets essential for determining veterans potentially exposed to

the Khamisiyah hazard area. The first is a registry of Gulf War veterans provided by DMDC and the second is a registry of daily unit locations during the war. The US Army Center for Health Promotion and Preventive Medicine combined USASCURR's troop locations with the 2000 modeling results to produce graphic representations of the hazard area for March 10 - 13, 1991. We refer to these graphics as the 2000 hazard area results, discussed in the following paragraphs.

2000 modeling estimates of the Pit demolitions that remain unchanged from 1997. These estimates remained the same for the 2000 modeling as the 1997 modeling of the Pit:

- the fraction of chemical warfare agent that entered the atmosphere through instant aerosolization, and the fraction of agent that spilled on wood and soil and evaporated over time;
- the chemical warfare agent evaporation rates from wood and soil;
- the initial ratio of sarin to cyclosarin (3:1) in the agent cloud; and
- agent purity (50 percent).

US Forces Possibly Exposed. The 1997 possibly exposed troop population was 98,910, compared to the 2000 potentially affected troop population of 100,923. Of that number, approximately 66,103 personnel were in both the 1997 and 2000 modeling potential hazard areas. Table 6 compares the daily differences of those troops possibly exposed in 1997 and 2000. Some soldiers were in the potential hazard areas on multiple days, so the total number of possible exposures in Table 6 is not the same number of individual soldiers possibly exposed.

Table 6. Comparing the daily differences in possible exposures

Day	1997 Possible Exposures	2000 Possible Exposures
March 10	18,814	45,226
March 11	79,058	61,480
March 12	3,287	4,192
March 13	1,638	0

Comparing the 2000 to the 1997 hazard areas. Figures 43 - 50 compare and contrast the results of the 2000 models' and the 1997 models' potential hazard areas by day for March 10-13, 1991. Figure 51 shows a smaller scale view of the first noticeable effects area for the 2000 model (applies to March 10 only).

Exposure thresholds. We used two exposure thresholds in both our 1997 and 2000 modeling efforts: first noticeable effects and general population limit. We also use these thresholds in our modeling to describe areas of potential exposure.

- **First noticeable effects (FNE).** This is the dosage expected to cause watery eyes, runny nose, tightness of chest, muscle twitching, sweating, and headache. Dosage is a cumulative exposure—a concentration of a chemical warfare agent over time. Increasingly higher dosages would produce vision impairment, incapacitation, and death.
- **General population limit (GPL).** The GPL represents the limit below which any member of the general population could be exposed (e.g., inhale) seven days a week, every week, for a lifetime, without experiencing any adverse health effects. Scientists performed exposure testing and computed the GPL from the results of that testing as modified by uncertainty factors such as short-term to long-term exposure, average human to sensitive human population, etc.

Areas of potential exposure.

- **Low-level Hazard area.** In 1997, we defined the area between the two thersholds as the low-level hazard area, an area where long-term effects from low-level exposure is possible, but symptoms would not exist.

- **Potential Hazard area.** In 2000, we refer to the area that meets or exceeds the GPL as the potential hazard area, an area where long-term effects from low-level exposure are possible and where acute symptoms might show.

Since the possible chemical warfare agent releases occured over a four-day period, our 2000 modeling used the GPL based on short-term exposures, as recommended by the US Army Center for Health Promotion and Preventive Medicine.[221] Figures 43-50 shows the maximum size of the potential hazard inside which personnel may have been exposed to a nerve agent concentration equal to or exceeding this GPL. Exposure of personnel outside this area did not reach hazardous levels. The dots on each graphic represent a unit's or part of a unit's location. The number of units is greater in 2000 because our knowledge of unit locations improved since 1997 due to the G3/S3 conferences.

Comparing each day's hazard area from March 10 to 13, 1991.

Day 1: March 10, 1991 (Figures 43 - 44)

The 1997 hazard area generally extended in a north-south direction. Based on the wind models, the hazard area extended south-southwest almost 300 kilometers into Saudi Arabia, east of the city of Hafir Al Batin. The 2000 hazard area is bifurcated due to slight differences in the wind patterns predicted by the meteorological models but generally follows a north-south axis. The hazard area extends approximately the same distance but to the northwest of Hafir Al Batin. However, the 2000 hazard area extended further to the east and south into Kuwait indicating possible exposure to US personnel not in the 1997 hazard area. Approximately 70 percent of the 321 kilograms of chemical warfare agent modeled in 2000 had been dispersed into the atmosphere by the end of Day 1.

Figure 43. 1997 Potential Hazard area for Day 1: March 10, 1991

Figure 44. 2000 Potential Hazard area for Day 1: March 10, 1991

Day 2: March 11, 1991 (Figures 45 - 46)

Both models' hazard areas have similar shapes. However, for the 2000 hazard area, the dispersion modeling assumed ultraviolet light from sunlight decayed the chemical warfare agent and thus reduced the size of the 2000 hazard area. The 1997 model did not consider sunlight effects on the hazard area, but we added these effects for the 2000 modeling at the recommendation of the 1997 peer review panel. Improved 2000 weather models reflected a shift in the wind to the south, which kept the 2000 hazard area mainly to the north and west of King Kalid Military City (KKMC), unlike the 1997 hazard area, which included KKMC and a much larger area to the west. Later nerve agent emissions evaporating from the soaked wood and soil in the Pit generated the four small hazard areas in the vicinity of Al Bussayyah and around Khamisiyah in the 2000 model. Approximately 89 percent of the 321 kilograms of the chemical warfare agent modeled in 2000 had dispersed into the atmosphere by the end of Day 2.

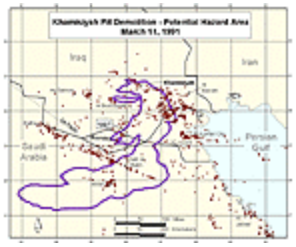

Figure 45. 1997 Potential Hazard area for Day 2: March 11, 1991

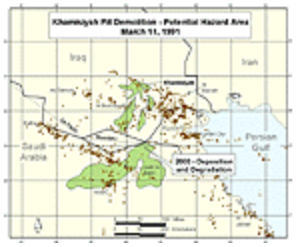

Figure 46. 2000 Potential Hazard area for Day 2: March 11, 1991

Day 3: March 12, 1991 (Figures 47 - 48)

The 1997 model shows the hazard area included some of the Euphrates River valley. The 2000 model hazard area is much smaller, due to incorporating two full days of ultraviolet light reaction in the dispersion model. By the third day the chemical warfare agent remaining in the atmosphere, soil, and wood was below the general population limit. Approximately 97 percent of the 321 kilograms of the chemical warfare agent modeled in 2000 had dispersed into the atmosphere by the end of Day 3.

Figure 47. 1997 Potential Hazard area for Day 3: March 12, 1991

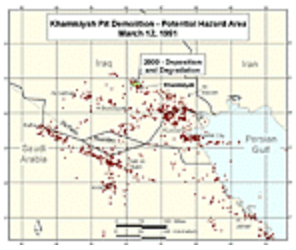

Figure 48. 2000 Potential Hazard area for Day 3: March 12, 1991

Day 4: March 13, 1991 (Figures 49 - 50)

A miniscule amount of agent evaporation generated a very small hazard area in both models. Approximately 100 percent of the 321 kilograms of the chemical warfare agent modeled in 2000 had dispersed into the atmosphere by the end of Day 4.

Figure 49. 1997 Potential Hazard area for Day 4: March 13, 1991

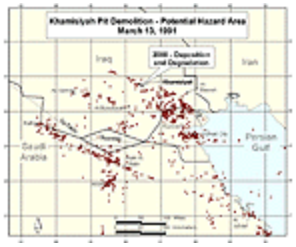

Figure 50. 2000 Potential Hazard area for Day 4: March 13, 1991

First noticeable effects 2000 hazard area: March 10, 1991 (Figure 51). The darker, smaller area on this figure is the 2000 FNE hazard area, and it is within the 2000 models' M8 alarm hazard area (that area in which the concentration of a chemical warfare agent would be enough to cause an M8 chemical agent detector to detect and sound an alarm for the presence of a nerve agent). Any person in the FNE hazard area would experience immediate visible signs of nerve agent exposure, such as tearing eyes or shortness of breath, and the M8 chemical agent detectors would have sounded a warning to go to a protective posture. No US units were in the first noticeable effects area in either the 1997 or 2000 models.

Figure 51. 2000 First noticeable effects and M8 alarm detection areas

Modeling Khamisiyah. Our modeling does not represent what really happened, but rather what may have happened, given what we know. Our modeling efforts over the past three years have suffered from the passage of time and a fundamental lack of measured data. We overcame single-model biases by using ensemble modeling techniques—using two or more models to predict the potential hazard area. Though we gained better data on the source term and also had improved models with which to work, we still did not have the accuracy and depth of data the models required to produce an accurate hazard area. Given the limited data we had, we estimated variables such as temperature, moisture, wind speed and direction, and the exact size of the source term. In the end, we again used conservative modeling parameters to attempt to identify the majority of those who may have been exposed in order to protect veterans' health. Servicemembers should consider their own particular health circumstances and evaluate the information in this narrative, given the uncertainties of modeling the Khamisiyah pit demolition activities. The individual veteran's health is our utmost concern, and we will leave no stone unturned to protect it.